KAIZEN MATHS

Lesson starter

Newton's Second Law: F = ma

F = ma Basics

Name _____

S1

A particle of mass 10 kg accelerates at 0.8 m s^{-2}. Find the resultant force acting on it.

S2

A resultant force of 4.8 N acts on a body of mass 12 kg. Find its acceleration.

S3

A body accelerates at 0.8 m s^{-2} when the resultant force is 4 N. Find its mass.

S4

Find the weight of an object of mass 12 kg. Take g = 9.8 m s^{-2}.

Show your working

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Exit ticket

Newton's Second Law: F = ma

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Name _____

E1

A particle of mass 2 kg accelerates at 2 m s^{-2}. Find the resultant force acting on it.

E2

A resultant force of 62.5 N acts on a body of mass 25 kg. Find its acceleration.

E3

A body accelerates at 2.4 m s^{-2} when the resultant force is 24 N. Find its mass.

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Show your working

Teacher answers & working

Newton's Second Law: F = ma · F = ma Basics

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S1

A particle of mass 10 kg accelerates at 0.8 m s^{-2}. Find the resultant force acting on it.

Answer:

8 N

Working:

Newton's second law states that resultant force equals mass multiplied by acceleration.

$F = ma$

$m = 10, a = 0.8$

$F = 10 \times 0.8$

$F = 8$ N

Resultant force = 8 N.

S2

A resultant force of 4.8 N acts on a body of mass 12 kg. Find its acceleration.

Answer:

0.4 m s^{-2}

Working:

Start with F = ma and rearrange to make acceleration the subject.

$F = ma$

Divide both sides by m.

$a = \frac{F}{m}$

$a = \frac{4.8}{12}$

$a = 0.4$ m s^{-2}

Acceleration = 0.4 m s^{-2}.

Teacher answers & working

Newton's Second Law: F = ma · F = ma Basics

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S3

A body accelerates at 0.8 m s^{-2} when the resultant force is 4 N. Find its mass.

Answer:

5 kg

Working:

Start with F = ma and rearrange to make mass the subject.

$F = ma$

Divide both sides by a.

$m = \frac{F}{a}$

$m = \frac{4}{0.8}$

$m = 5 \text{ kg}$

Mass = 5 kg.

S4

Find the weight of an object of mass 12 kg. Take g = 9.8 m s^{-2}.

Answer:

117.6 N

Working:

Weight is the gravitational force on a mass.

$W = mg$

$m = 12, g = 9.8$

$W = 12 \times 9.8$

$W = 117.6 \text{ N}$

Weight = 117.6 N.

E1

A particle of mass 2 kg accelerates at 2 m s^{-2}. Find the resultant force acting on it.

Answer:

4 N

Working:

Newton's second law states that resultant force equals mass multiplied by acceleration.

$F = ma$

$m = 2, a = 2$

$F = 2 \times 2$

$F = 4 \text{ N}$

Resultant force = 4 N.

Teacher answers & working

Newton's Second Law: F = ma · F = ma Basics

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E2

A resultant force of 62.5 N acts on a body of mass 25 kg. Find its acceleration.

Answer:

$2.5\ \text{m s}^{-2}$

Working:

Start with F = ma and rearrange to make acceleration the subject.

$F = ma$

Divide both sides by m.

$a = \frac{F}{m}$

$a = \frac{62.5}{25}$

$a = 2.5\ \text{m s}^{-2}$

Acceleration = 2.5 m s^{-2}.

E3

A body accelerates at 2.4 m s^{-2} when the resultant force is 24 N. Find its mass.

Answer:

$10\ \text{kg}$

Working:

Start with F = ma and rearrange to make mass the subject.

$F = ma$

Divide both sides by a.

$m = \frac{F}{a}$

$m = \frac{24}{2.4}$

$m = 10\ \text{kg}$

Mass = 10 kg.